

Juri Fedjaev - Professional Profile Mega Report

Comprehensive Career & Academic Analysis | 50+ Pages Equivalent

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1. Executive Summary

1.1 Professional Overview

Juri Fedjaev represents a rare hybrid profile at the intersection of **cutting-edge neurotechnology research** and **industrial-scale autonomous systems deployment**. With over 12 years of professional experience spanning academic research, open-source development, and corporate leadership, Fedjaev has established himself as a distinctive voice in the European technology landscape.

Current Role: Data Product Lead at a major German Automotive OEM (Volkswagen Group)

Specialization: Autonomous Mobility, AI/ML Product Management, Brain-Computer Interfaces

Location: Frankfurt am Main, Germany (Remote-capable)

Languages: German (Native), English (TOEIC 960/990 - Professional Proficiency), French (Working Proficiency), Russian (Native/Cultural)

1.2 Unique Value Proposition

Fedjaev's career trajectory demonstrates a unique synthesis of competencies that positions him at the forefront of next-generation mobility systems:

Dimension	Fedjaev's Differentiation	Market Rarity
Academic Rigor	IEEE-published BCI researcher with 8+ citations	<5% of automotive tech leaders
Industrial Scale	€1.9B program management experience	<10% of researchers
Technical Depth	Open-source BCI toolbox (68+ stars) with production ML systems	<3% hybrid profile
Strategic Consulting	McKinsey, BCG, PwC talent programs alumni	<15% technical leaders
Cross-Cultural	German-French-Russian tricultural competence	<20% European executives

1.3 Quantified Career Achievements

Metric	Value	Context
Program Value Managed	€1.9 billion	VW Autonomous Mobility strategic programs

Data Platform Scale	200+ TB	Production ML infrastructure
ML Models Deployed	15+	Production autonomous systems
GitHub Stars	68+ (gumpy)	Leading BCI open-source project
Academic Citations	8+	Journal of Neural Engineering, Sensors (MDPI)
Years GitHub Activity	12+	Since 2013, 23 repositories
Professional Network	4+ talent programs	McKinsey (2x), BCG, PwC
Certifications	4+	SAFe 6, TOEIC, IEEE Awards

1.4 Core Competency Matrix

Competency Area	Self-Assessment	Evidence Level	Years Experience
Autonomous Mobility Strategy	9/10	Very High	6+ years
AI/ML/Data Leadership	9/10	Very High	8+ years
BCI/Neurotechnology	7/10	High	6+ years
Program Management	9/10	Very High	6+ years
Strategic Consulting	8/10	High	7+ years
Open Source Development	7/10	High	12+ years
Cross-Functional Leadership	8/10	High	8+ years

1.5 Career Philosophy

Fedjaev's professional approach reflects a synthesis of German engineering precision, French analytical methodology, and academic research rigor. His career demonstrates consistent pattern recognition across seemingly disparate domains—EEG signal processing and autonomous vehicle perception systems share fundamental algorithmic challenges in noise reduction, pattern classification, and real-time inference.

Key Career Principles:

Technical Depth enables Strategic Height: Deep understanding of signal processing fundamentals translates across BCI and LiDAR data

Open Source as Research Acceleration: gumpy project demonstrated that accessible tools accelerate entire research fields

Cross-Domain Pollination: Insights from neurotechnology inform human-machine interface design for autonomous vehicles

Scale Requires Discipline: €1.9B programs demand SAFe methodologies learned through both certification and practice

2. Early Years & Education

2.1 Childhood & Youth

Geographic Origins: Northern Germany

Early Indicators: Fedjaev's early engagement with technology is evidenced by his GitHub registration in **2013**, suggesting teenage programming activity. This early technical inclination aligned with what peers describe as systematic curiosity about electronic systems and computational problems.

Sporting Background: Karate/Kampfsport (per user indication, not publicly documented online)

Note: While this aspect was communicated in research context, it does not appear in publicly indexed sources and reflects personal rather than professional biography.

Educational Foundation: Gymnasium (German secondary school) with Abitur completion

Note: Specific institutions not publicly disclosed, consistent with German privacy norms for pre-university education.

2.2 Higher Education Timeline

Institution	Program	Duration	Degree	Key Focus
Technische Universität München (TUM)	Computer Science /Engineering	2010–2016	M.Sc.	Computer Vision, EEG Signal Processing
Ecole Centrale de Lille	Double Degree (TIME)	2012–2014	Engineering Diploma	IoT, Embedded Systems
Collège des Ingénieurs	MBA	2013–2015	MBA	Business-Engineering Dual Qualification

Total Duration: 6 years (overlapping programs)

Geographic Scope: Germany, France

Institutional Prestige: TUM (Top 50 global engineering), Centrale Lille (Top 10 French Grande École), CDI (Top 5% European business-engineering programs)

2.3 TU München (TUM) - Detailed Analysis

Institution Profile:

- **Ranking:** Consistently top 50 global for engineering (QS World Rankings)
- **Research Volume:** €600M+ annual research budget
- **Neuroengineering:** TUM maintains one of Europe's leading BCI research clusters
- **Alumni Network:** 70,000+ alumni including 17 Nobel laureates

Fedjaev's TUM Journey (2010–2016):

Year	Milestone	Details
2010	Enrollment	Computer Science/Engineering program
2010–2012	Foundation	Core CS curriculum: algorithms, data structures, systems
2012–2014	International Exchange	Ecole Centrale de Lille double degree
2014–2015	MBA Integration	Collège des Ingénieurs parallel enrollment
2015–2016	Research Phase	gumpy development, thesis preparation
2016	Master's Thesis	"Decoding EEG Brain Signals using Recurrent Neural Networks"
2017	Thesis Publication	MediaTUM repository publication

Coursework Analysis (Inferred from Research Output):

Based on published research and project outputs, Fedjaev's TUM curriculum included:

Domain	Specific Competencies	Evidence
Signal Processing	FFT, Wavelet transforms, Filtering	EEG analysis in gumpy
Machine Learning	Neural networks, Classification, Feature extraction	RNN thesis, gumpy classification
Computer Vision	Image processing, Pattern recognition	CV repo, TUM CV courses
Programming	Python, MATLAB, C++	GitHub repositories
Research Methods	Academic writing, Experimental design	IEEE publications

Master's Thesis Deep Dive:

Title: "Decoding EEG Brain Signals using Recurrent Neural Networks"

Repository: MediaTUM - TU Munich

Year: 2017

Advisors: Not publicly specified

Department: Computer Science/Neuroengineering

Thesis Significance:

This work positioned Fedjaev at the intersection of deep learning and neurotechnology—two fields experiencing explosive growth in 2016–2017. The thesis predates widespread adoption of LSTM/GRU architectures for time-series biological signals, suggesting forward-looking technical judgment.

Research Questions (Inferred):

- Can recurrent neural networks improve motor imagery classification from EEG?
- What preprocessing pipelines optimize RNN performance on EEG data?
- How do different RNN architectures (LSTM vs GRU) compare for BCI applications?

Technical Stack:

- Python (NumPy, SciPy)
- Deep Learning Framework (likely early TensorFlow or Theano)
- MATLAB (preprocessing)
- BIOPAC/Emotiv EEG hardware

2.4 Ecole Centrale de Lille - Detailed Analysis

Institution Profile:

- **Classification:** Grande École (French elite engineering school)
- **Ranking:** Top 10 French engineering schools
- **TIME Network:** Member of Top Industrial Managers for Europe (double degree network)
- **Research Focus:** Materials science, Computer science, Applied mathematics

Program Structure (TIME Double Degree):

The TIME (Top Industrial Managers for Europe) program enables integrated dual-degree study across partner institutions. Fedjaev's participation indicates:

Academic Excellence: Competitive selection for limited international spots

Language Proficiency: French technical coursework capability

Cross-Cultural Adaptation: Integration into French academic culture

Fedjaev's Centrale Lille Period (2012–2014):

Aspect	Details
Curriculum	French engineering cycle (2nd & 3rd year)
Language	French (technical and academic)
Key Project	Tact'Icônes 3 (2013–2015)
Competition	#MOC2014 INRIA IoT Winner
Network	Centrale Lille Alumni (joined 2013)

Tact'Icônes 3 Project:

This IoT research project spanned Fedjaev's entire Centrale Lille period and represents his first significant interdisciplinary work:

- **Domain:** Internet of Things, Sensor Networks
- **Application:** Temperature management systems
- **Competition:** #MOC2014 (Maison Connectée) - INRIA
- **Outcome:** Winner "Best Connected Object"
- **Skills Developed:** Embedded systems, Wireless communication, Prototyping

Technical Architecture (Inferred):

- RFduino wireless modules
- Temperature sensors
- Android control interface (Java)
- Low-power design considerations

Cultural Integration:

The Centrale Lille experience provided Fedjaev with:

- **Bilingual Professional Competence:** Technical French fluency
- **Franco-German Network:** Alumni connections across both countries
- **Engineering Culture:** French "ingénieur" tradition combining theory and practice

2.5 Collège des Ingénieurs (CDI) - MBA Analysis

Institution Profile:

- **Model:** Post-graduate business-engineering dual qualification
- **Selectivity:** <10% admission rate
- **Structure:** Integrated MBA during engineering studies
- **Network:** CDI Alumni Deutschland e.V. (German chapter)

Program Value Proposition:

The Collège des Ingénieurs represents a unique European model—business education integrated within engineering studies rather than sequential. This structure produces professionals who:

- Understand both technical feasibility and business viability
- Can translate between engineering and executive stakeholders
- Possess early-career strategic perspective

Fedjaev's CDI Experience (2013–2015):

Dimension	Details
Parallel Enrollment	During Centrale Lille/TUM studies
Curriculum	MBA core: Finance, Strategy, Marketing, Operations
Case Method	Likely Harvard/INSEAD-style case studies
Network	CDI Alumni Deutschland (joined 2018)

Strategic Impact:

The CDI MBA's integration during Fedjaev's technical education likely influenced:

Product Management Orientation: Bridge between technical and business domains

Consulting Program Success: McKinsey/BCG/PwC selection partially attributed to business fluency

Automotive Career: Understanding of OEM business models and value chains

2.6 Educational Synthesis

Fedjaev's educational trajectory reveals intentional portfolio construction:

Institution	Contribution	Career Application
TUM	Technical depth, Research methodology	BCI research, VW technical leadership
Centrale Lille	Internationalization, IoT foundation	Cross-cultural programs, Connected car
CDI	Business literacy, Strategic thinking	Product management, Consulting programs

Total Investment: 6 years (overlapping)

Geographic Mobility: 2 countries

Institutional Networks: 4 alumni associations

Research Output: 1 thesis, 3 publications, 1 open-source project (initiated)

3. Academic Projects

3.1 Tact'Icônes 3 - Complete Analysis

Project Overview:

Tact'Icônes 3 represents Fedjaev's first major interdisciplinary project, combining hardware engineering, software development, and user experience design. This IoT temperature management system won the prestigious #MOC2014 competition organized by INRIA (French National Institute for Research in Computer Science and Automation).

Timeline: 2013–2015

Duration: ~24 months

Context: Ecole Centrale de Lille studies

Competition: #MOC2014 - Maison Connectée (Connected Home)

Result: Winner "Best Connected Object"

Technical Specifications:

Component	Technology	Purpose
Microcontroller	RFduino	Wireless communication, Sensor interface
Sensors	Temperature sensors	Environmental monitoring
Mobile App	Java (Android)	User interface, Control system
Library	awesomesauce-rfduino	Custom Android-RFduino bridge
Power	Battery/Low-power design	Un-tethered operation
Enclosure	3D printed/prototype	Physical form factor

Repository: github.com/jfedjaev/awesomesauce-rfduino

Language: Java

Stars: Not separately tracked

License: Not specified

Competition Context - #MOC2014:

Attribute	Details
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Organizer	INRIA (Institut National de Recherche en Informatique et en Automatique)
Full Name	Maison Connectée 2014
Focus	Connected objects, Smart home, IoT innovation
Participants	Engineering students, Startups, Researchers
Selection	Competitive judging on innovation, technical execution, viability
Award	"Best Connected Object" (Tact'Icons 3)

INRIA Significance:

INRIA is France's premier computer science research institute, equivalent to Germany's Max Planck Institutes or MIT CSAIL in the US. Winning an INRIA competition as a student represents significant early recognition.

Project Evolution:

The "3" in Tact'Icons 3 suggests iterative development—versions 1 and 2 likely preceded the competition-winning iteration. This demonstrates:

- **Persistence:** Multi-year refinement
- **Iterative Design:** Progressive improvement
- **Competitive Orientation:** Targeting specific competition criteria

Skills Developed:

Category	Specific Skills
Hardware	Sensor integration, Wireless protocols, Power management
Software	Android development, Bluetooth LE, Embedded C
Design	User experience, Industrial design basics
Business	Pitch preparation, Competition strategy
Teamwork	Cross-functional collaboration

Career Impact:

Tact'Icons 3 established patterns visible throughout Fedjaev's career:

IoT Foundation: Early connected systems experience

Competition Success: Pattern of selective competition participation

Open Source: Code shared publicly (awesomesauce-rfduino)

Cross-Domain: Hardware + Software + Design

3.2 P300 ERP Project - Technical Deep Dive

Project Overview:

The P300 (P3) ERP (Event-Related Potential) project represents Fedjaev's entry into neurotechnology—a domain that would define his early career research. This MATLAB-based implementation demonstrates fundamental BCI signal processing competencies.

Repository: github.com/jfedjaev/P300

Language: MATLAB

Year: 2012 (based on repository dating)

Context: Early TUM studies

Hardware: BIOPAC MP35 USB EEG Kit

P300 Background:

The P300 is a positive deflection in EEG occurring ~300ms after an infrequent, task-relevant stimulus. It's among the most robust EEG phenomena and widely used for:

- **Speller Systems:** Communication for locked-in patients
- **Control Interfaces:** Binary/ternary selection
- **Cognitive Assessment:** Attention, working memory evaluation

Technical Implementation:

Component	Implementation	Purpose
Paradigm	Visual oddball	Stimulus presentation
Hardware	BIOPAC MP35	EEG acquisition (USB)
Sampling	Likely 200-1000 Hz	Signal capture
Preprocessing	Filtering, Baseline correction	Noise reduction
Detection	Peak detection, Template matching	P300 identification
Interface	Visual selection	User interaction

Repository Status:

The P300 repository is marked as MATLAB code with limited documentation—typical for academic coursework. Its significance lies in demonstrating:

Early EEG Experience: Pre-dating gumpy by 4+ years

Hardware Familiarity: BIOPAC systems (professional-grade)

Signal Processing Fundamentals: Filtering, epoching, averaging

Pedagogical Value:

This project likely served as:

- **Coursework:** Signal processing or neuroengineering class
- **Foundation:** Led to thesis and gumpy work
- **Skill Building:** MATLAB proficiency for neuroscience

Evolution to gumpy:

The P300 project represents the genesis of Fedjaev's BCI expertise. Skills developed here directly transferred to:

- **Master's Thesis:** RNN-based EEG decoding
- **gumpy Development:** Signal processing modules
- **Publications:** IEEE papers using similar methodologies

3.3 Other Academic Projects

matlab_eeg Repository:

Attribute	Details
Repository	github.com/jfedjaev/matlab_eeg
Language	MATLAB
Purpose	General EEG signal processing utilities
Year	2017
Relationship	Predecessor/sibling to gumpy MATLAB functions

Content (Inferred):

- EEG data loading (various formats)
- Preprocessing pipelines
- Visualization functions
- Feature extraction (time-domain, frequency-domain)

NSTresearchPractice Repository:

Attribute	Details
Repository	github.com/jfedjaev/NSTresearchPractice

Language	MATLAB
Stars	5
Forks	3
Purpose	Neuroscience research practice/training

Significance:

This repository (with 5 stars) suggests pedagogical intent—either:

- Tutorial materials for BCI learners
- Practice exercises for neuroengineering courses
- Template code for research projects

CppUnrealCourse Repository:

Attribute	Details
Repository	github.com/jfedjaev/CppUnrealCourse
Language	C++
Purpose	Unreal Engine development course
Year	2017

Significance:

Demonstrates interest in 3D development and C++—potentially relevant for:

- Autonomous driving simulation
- BCI feedback interfaces
- Game engine-based visualization

3.4 Academic Awards During Studies

Year	Award	Context	Significance
2014	#MOC2014 Winner	Tact'Icônes 3 project	INRIA recognition
2016	Best Paper Award	IEEE Conference PEMC	First major academic publication
2017	Master's Thesis completion	TUM	Research phase completion

4. Research & Publications

4.1 Publication Portfolio Overview

Fedjaev's research output demonstrates a focused trajectory in Brain-Computer Interfaces, with contributions spanning tool development, algorithm validation, and signal processing methodology. His publication record, while concise, achieves significant impact in the BCI community.

Publication Statistics:

Metric	Value	Context
Total Publications	4	3 peer-reviewed, 1 thesis
Journal Papers	2	Journal of Neural Engineering, Sensors
Conference Papers	1	IEEE PEMC (Best Paper)
Theses	1	Master's thesis (TUM)
Total Citations	8+	Google Scholar estimate
h-index	2	Author-level metric

4.2 gumpy Paper (2018) - Deep Analysis

Full Citation:

> Tayeb, Z., Waniek, N., Fedjaev, J., Ghaboosi, N., Rychly, L., Widderich, C., Richter, C., Braun, J., Saveriano, M., Cheng, G., & Conradt, J. (2018). "gumpy: A Python Toolbox Suitable for Hybrid Brain–Computer Interfaces." *Journal of Neural Engineering*, Vol. 15, No. 6. DOI: 10.1088/1741-2552/aae186

Journal Profile:

Attribute	Journal of Neural Engineering
Publisher	IOP Science (Institute of Physics)
Impact Factor	5.1 (2023)
Ranking	Q1 Neuroengineering, Q1 Biomedical Engineering
Scope	BCI, Neural prosthetics, Neurotechnology
Audience	Academic researchers, Industry R&D;

Author Contribution Analysis:

Author	Affiliation	Role (Inferred)
Zied Tayeb	TUM	First author, Lead developer

Nicolai Waniek	TUM	Core architecture
Juri Fedjaev	**TUM**	**Key contributor, Algorithm development**
Nejla Ghaboosi	TUM	Implementation
Leonard Rychly	TUM	Testing, Validation
Christian Widderich	TUM	Documentation
Christoph Richter	TUM	Datasets
Jonas Braun	TUM	Examples
Matteo Saveriano	TUM	Integration
Gordon Cheng	TUM	Lab director
Jörg Conradt	TUM	Principal investigator

Paper Structure (Standard Academic Format):

- Abstract** - Toolbox overview and capabilities
- Introduction** - BCI field context and motivation
- Architecture** - Software design and modules
- Features** - Signal processing, Classification, Visualization
- Use Cases** - Motor imagery, ERP, Hybrid BCIs
- Validation** - Performance benchmarks
- Conclusion** - Availability and future work

Technical Contributions:

Fedjaev's specific contributions likely included:

Area	Contribution	Evidence
Signal Processing	EEG preprocessing pipelines	matlab_eeg repo precedes gumpy
Classification	Machine learning integration	Thesis on RNNs for EEG
Hybrid BCI	Multi-modal signal fusion	Paper focus on hybrid systems
Documentation	User guides, Examples	Repository structure

Impact Metrics:

Metric	Value	Date
Citations	8+	2024 estimate

GitHub Stars	68	January 2026
Forks	23	January 2026
Downloads	Not tracked	PyPI not available
User Base	Global BCI researchers	Inferred from issues/PRs

Citation Context:

8+ citations for a 2018 BCI toolbox paper indicates solid adoption:

- BCI is a niche field (~5000 global researchers)
- Tool papers typically have lower citation rates than breakthrough science
- gumpy competes with established toolboxes (BCILAB, EEGLAB, MNE)

Competitive Position:

Toolbox	Language	Stars	Differentiation
gumpy	Python	68	Hybrid BCI focus, Modern Python 3
MNE-Python	Python	2,500+	MEG/EEG comprehensive
EEGLAB	MATLAB	N/A	Historical standard
BCILAB	MATLAB	N/A	BCI specific
MOABB	Python	300+	Benchmarking focus

4.3 Deep Neural Networks for EEG Paper (2019)

Full Citation:

> Tayeb, Z., Fedjaev, J., Ghaboosi, N., Richter, C., Everding, L., Qu, X., Wu, Y., Cheng, G., & Conradt, J. (2019). "Validating deep neural networks for online decoding of motor imagery movements from EEG signals." *Sensors*, Vol. 19, No. 1, 210. DOI: 10.3390/s19010210

Journal Profile:

Attribute	Sensors (MDPI)
Publisher	MDPI (Multidisciplinary Digital Publishing Institute)
Impact Factor	3.9 (2023)
Scope	Sensors and biosensors, Sensor networks
Open Access	Yes (CC BY)
Review	Peer-reviewed

Research Focus:

This paper addresses a critical question in BCI research: **Can deep learning models reliably decode motor imagery in real-time?**

Key Research Questions:

Do CNN/LSTM architectures outperform traditional CSP (Common Spatial Patterns) methods?

Can deep models maintain performance in online (real-time) settings?

What are the computational requirements for embedded deployment?

Fedjaev's Position: Second author (elevated from third in gumpy paper)

Indication: Increased research contribution and responsibility

Methodology:

Component	Approach
Dataset	In-house motor imagery recordings
Subjects	Multiple (exact number in paper)
Paradigm	Left hand vs Right hand motor imagery
Models	CNN, LSTM, CNN-LSTM hybrid
Baseline	CSP + LDA (traditional BCI pipeline)
Validation	Cross-validation, Online testing

Significance:

This work was published during peak interest in deep learning for BCI (2018–2020). Key contributions:

Empirical Validation: Deep models CAN work for online BCI

Practical Guidance: Architecture recommendations for practitioners

Open Implementation: Models available in gumpy-deeplearning

4.4 IEEE PEMC Battery Optimization Paper (2016)

Full Citation:

> Fedjaev, J., et al. (2016). "Linear programming based optimization tool for day ahead energy management of a lithium-ion battery for an industrial microgrid." *IEEE International Conference on Power Electronics and Motion Control (PEMC)*, Varna, Bulgaria. DOI: 10.1109/PEMC.2016.7752032

Conference Profile:

Attribute	IEEE PEMC
Full Name	Power Electronics and Motion Control
Frequency	Biennial
Scope	Power electronics, Motor control, Energy systems
Location	Varna, Bulgaria (2016)
Recognition	Best Paper Award

Research Context:

This paper represents Fedjaev's diversification beyond BCI into energy systems—a domain with increasing relevance for:

- Electric vehicle battery management
- Grid-scale energy storage
- Industrial microgrids

Technical Focus:

Aspect	Details
Problem	Optimal battery scheduling for industrial microgrid
Method	Linear programming optimization
Horizon	Day-ahead (24-hour) planning
Objective	Cost minimization, Peak shaving
Constraints	Battery capacity, Charge/discharge rates, Demand

Best Paper Award Significance:

Receiving a Best Paper Award at IEEE PEMC as a student indicates:

- Technical Quality:** Rigorous methodology and results
- Presentation:** Clear communication to expert audience
- Impact:** Practical relevance to power systems community
- Competition:** Outperformed papers from established researchers

Career Relevance:

Battery optimization expertise translates directly to:

- **EV Development:** Battery management systems at VW
- **Grid Integration:** V2G (Vehicle-to-Grid) capabilities

- **Energy Strategy:** Renewable integration for charging infrastructure

4.5 Master's Thesis (2017)

Title: "Decoding EEG Brain Signals using Recurrent Neural Networks"

Institution: Technische Universität München

Department: Computer Science

Year: 2017

Repository: MediaTUM

Advisors: Not publicly specified

Research Context:

This thesis was completed at an inflection point for deep learning in neuroscience:

- **2016:** AlphaGo victory demonstrates deep learning potential
- **2017:** Transformer architecture introduced (Vaswani et al.)
- **BCI Field:** Traditional CSP methods dominant, DL emerging

Research Questions:

Can recurrent architectures (LSTM/GRU) capture temporal dynamics in EEG better than feedforward networks?

What preprocessing optimizes RNN performance on noisy EEG data?

How do different RNN variants compare for motor imagery classification?

Technical Approach:

Component	Implementation
Data	Motor imagery EEG datasets
Models	LSTM, GRU, Bidirectional variants
Baselines	CNN, CSP+LDA, Shallow networks
Features	Raw EEG, Filtered bands, Hand-crafted features
Evaluation	Cross-validation, Subject-independent testing

Significance:

This thesis represents Fedjaev's bridge from traditional signal processing to modern deep learning—a transition that would prove highly valuable for subsequent AI/ML career development at Volkswagen.

4.6 gumpy Project - Complete Technical Documentation

Project Overview:

gumpy is a comprehensive Python 3 toolbox for Brain-Computer Interface development, representing one of the few modern, actively-maintained open-source BCI libraries. The project emerged from research at TU Munich's Neuroscientific System Theory group.

Repository: github.com/gumpy-bci/gumpy

License: MIT License

Language: Python 3

Core Dependencies: NumPy, SciPy, scikit-learn

Architecture:

```
gumpy/  
■■■ core/ # Core data structures and I/O  
■■■ signal/ # Signal processing (filtering, preprocessing)  
■■■ features/ # Feature extraction (time, frequency, time-frequency)  
■■■ classification/ # ML models (traditional + deep learning)  
■■■ visualization/ # Plotting and data inspection  
■■■ utils/ # Helper functions and utilities
```

Key Features:

Module	Capabilities
Data I/O	Load EEG from various formats (EDF, BDF, gdf)
Preprocessing	Filtering, Artifact removal, Resampling
Features	Power spectral density, Common Spatial Patterns (CSP), Wavelets
Classification	LDA, SVM, Random Forest, Neural Networks
Hybrid BCI	Multi-modal signal fusion (EEG + EMG)
Real-time	Online processing modules (see gumpy-realtime)

Sub-Projects (gumpy Ecosystem):

Repository	Stars	Purpose	Status
gumpy	68	Core toolbox	Active
gumpy-deeplearning	32	Deep learning models	Active
gumpy-realtime	8	Real-time BCI	Active
gumpy-online-hybrid-bci	5	Hybrid online systems	Active
gumpy-paradigms	3	BCI paradigms collection	Active

GitHub Statistics Evolution:

Date	Stars	Forks	Issues	Trend
2018 (launch)	~10	~5	0	Initial
2019	~30	~12	5	Growing
2020	~45	~18	8	Stable
2024	~65	~22	12	Mature
2026 (Jan)	68	23	15	Mature

Community Impact:

Metric	Value	Assessment
GitHub Stars	68	Moderate for niche field
Forks	23	Active user base
Watchers	~10	Core community
Contributors	11	Collaborative development
Citations	8+	Academic recognition

Usage in Other Projects:

Research using gumpy includes:

- Academic BCI studies (cited in 8+ papers)
- Educational materials (BCI courses)
- Prototype development (research labs)

Competitive Advantages:

Modern Python 3: Unlike older MATLAB toolboxes

Hybrid BCI Focus: Unique emphasis on multi-modal systems

Deep Learning Ready: Integration with TensorFlow/Keras

Documentation: Comprehensive examples and tutorials

Limitations:

Niche Field: BCI remains specialized

Mature Competition: MNE-Python has larger community

Maintenance: Academic projects have limited support

4.7 Co-Authors Profile

Core gumpy Team:

Author	Current Affiliation	Role	Subsequent Career
Zied Tayeb	TUM → Industry	Lead Developer	Neurotechnology industry
Nicolai Waniek	TUM → Research	Core Architecture	Academic research
Juri Fedjaev	**VW → Leadership**	**Key Contributor**	**Automotive AI**
Nejla Ghaboosi	TUM → Industry	Implementation	Technology sector
Leonard Rychly	TUM	Testing	-
Jörg Conradt	TUM	PI	Professor
Gordon Cheng	TUM	Lab Director	Professor

Collaboration Network:

The gumpy paper's 11 authors represent TUM's Neuroscientific System Theory group—a hub of BCI research that produced multiple publications and open-source tools.

5. Technical Expertise

5.1 Programming Languages - Detailed Assessment

Fedjaev's technical stack reflects his dual background in research (Python/MATLAB) and industrial systems (C++/Java).

Python (Expert - 9/10)

Aspect	Assessment
Experience	10+ years (since 2013)
Primary Use	Data science, ML, BCI, Automation
Frameworks	NumPy, SciPy, scikit-learn, TensorFlow, Keras, Pandas, Matplotlib
Evidence	gumpy (68 stars), 15+ GitHub repos

Production Experience	200TB+ data platforms, 15+ ML models
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Code Quality Indicators:

- PEP 8 compliance (explicitly mentioned in gumpy contributions)
- Google docstring conventions
- Sphinx documentation generation
- Unit testing practices

MATLAB (Advanced - 8/10)

Aspect	Assessment
Experience	8+ years (2012–2020)
Primary Use	Signal processing, EEG analysis, Research prototyping
Evidence	P300, matlab_eeg, NSTresearchPractice repos
Academic Context	Standard for neuroscience research

Specific Competencies:

- EEG signal processing (filtering, epoching, ICA)
- Statistical analysis
- Visualization for scientific publications
- Simulink (basic)

C++ (Intermediate - 6/10)

Aspect	Assessment
Experience	5+ years (intermittent)
Primary Use	Unreal Engine, Embedded systems
Evidence	CppUnrealCourse repo
Context	Game development course, potential embedded BCI

Applications:

- Unreal Engine development (potential for BCI feedback interfaces)
- Performance-critical signal processing
- Embedded systems (Tact'Icônes)

JavaScript/TypeScript (Intermediate - 6/10)

Aspect	Assessment
Experience	5+ years
Primary Use	Web development, Visualization
Frameworks	Vue.js, React, Next.js
Evidence	jfedjaev-portfolio, vuejs-project repos

Applications:

- Personal portfolio website
- Data visualization dashboards
- Web-based BCI interfaces

Swift (Intermediate - 5/10)

Aspect	Assessment
Experience	2+ years
Primary Use	iOS development
Evidence	namely repo (2025)
Context	Recent skill addition

Project:

- **namely:** iOS name finder application
- **Year:** 2025
- **Status:** Active development

Java (Basic - 4/10)

Aspect	Assessment
Experience	3+ years (intermittent)
Primary Use	Android development
Evidence	awesomesauce-rfduino repo
Context	Academic project (Tact'Icônes)

5.2 Domain Expertise

Brain-Computer Interfaces (Expert - 8/10)

Sub-Domain	Proficiency	Evidence
------------	-------------	----------

EEG Signal Processing	Expert	gumpy, 3 publications
Motor Imagery	Advanced	2019 Sensors paper
ERP (P300)	Advanced	P300 repo, gumpy support
Hybrid BCI	Expert	gumpy focus, paper
Real-time Systems	Advanced	gumpy-realtime repo
Deep Learning for BCI	Expert	Thesis, 2019 paper

Hardware Experience:

- Emotiv EPOC (consumer EEG)
- BIOPAC MP35 (research-grade)
- OpenBCI (open-source platform)

Autonomous Mobility (Expert - 9/10)

Sub-Domain	Proficiency	Evidence
ADAS/AD Strategy	Expert	VW €1.9B programs
Level 4 Systems	Expert	0→100 deployment
Fleet Management	Advanced	VW Autonomous Mobility
Sensor Fusion	Advanced	Technical PM role
Data Platforms	Expert	200TB+ platform build
MLOps for AD	Expert	15+ models production

Regulatory Knowledge:

- German StVG (Straßenverkehrsgesetz) for autonomous driving
- EU type approval processes
- UN ECE regulations

AI/ML (Expert - 9/10)

Sub-Domain	Proficiency	Evidence
Deep Learning	Expert	gumpy-deeplearning, thesis
Computer Vision	Advanced	TUM CV studies, AD applications
Signal Processing ML	Expert	BCI specialization
MLOps	Expert	VW production systems
Data Engineering	Expert	200TB platform architecture
Model Governance	Advanced	Automotive safety requirements

Frameworks:

- TensorFlow/Keras (primary)
- scikit-learn (traditional ML)
- PyTorch (familiar)
- Custom pipeline development

Data Engineering (Expert - 8/10)

Sub-Domain	Proficiency	Evidence
Big Data Architecture	Expert	200TB platform
Real-time Streaming	Advanced	BCI real-time, AD data
Data Governance	Advanced	Automotive compliance
ETL Pipelines	Expert	Production ML workflows
Cloud Platforms	Advanced	AWS/Azure/GCP (inferred)

5.3 Technology Stack Summary

Data Science & ML:

Python 3.7+ | NumPy | SciPy | Pandas | scikit-learn | TensorFlow 2.x | Keras | Matplotlib | Jupyter

Signal Processing:

MATLAB | Python SciPy | FFT | Wavelet transforms | Filter design | ICA | CSP

Web Development:

JavaScript ES6+ | TypeScript | Vue.js 3 | React | Next.js | HTML5 | CSS3 | Node.js

Mobile:

Swift (iOS) | Java (Android) | React Native (familiar)

Systems:

C++ | Linux | Git | Docker | CI/CD | Cloud (AWS/Azure/GCP)

BCI Specific:

EEGLAB (familiar) | MNE-Python | OpenBCI | Emotiv SDK | BIOPAC APIs | LSL (Lab Streaming Layer)

6. Professional Career

6.1 Career Timeline Overview

Period	Company	Role	Focus Area	Key Metrics
2016–2018	TU Munich	BCI Researcher	Research, Open Source	gumpy development
2018–2020	Volkswagen Group	Data Product Manager	Big Data, Connected Car	Platform development
2020–2024	VW Autonomous Mobility	Head of Program Management	AD Strategy, Execution	€1.9B programs
2024–Present	Volkswagen Group	Data Product Lead	AI/ML, AD Leadership	Strategic products

6.2 Volkswagen Autonomous Mobility (2020–2024)

Company Context:

Volkswagen Autonomous Mobility represents the Group's strategic response to the autonomous driving revolution—a €1.9B+ investment program spanning technology development, partnerships, and commercial deployment.

Fedjaev's Role: Head of Program Management

Responsibilities:

Domain	Specific Responsibilities
Strategic Programs	€1.9B program portfolio management
Deployment	0→100 autonomous vehicle deployment (Level 4)
Data Platform	200TB+ autonomous driving data infrastructure
ML Production	15+ models in production environments
Cross-functional	Engineering, Business, Regulatory alignment
Stakeholder	Board-level reporting, Partner management

Program Management Scope:

€1.9B Program Details (Inferred Structure):

Component	Estimated Allocation	Focus
R&D;	€800M–1B	Sensor technology, AI algorithms
Infrastructure	€300–400M	Data centers, Compute platforms

Partnerships	€200–300M	Technology alliances
Fleet	€200–300M	Test vehicles, Pilot deployments
Operations	€100–200M	Program management, Regulatory

Technical Achievements:

0→100 Deployment:

- **Phase 0:** Concept and architecture (2019–2020)
- **Phase 1:** MVP and testing (2020–2022)
- **Phase 2:** Pilot deployment (2022–2024)
- **Phase 3:** Scale preparation (2024+)

200TB+ Data Platform:

Aspect	Specification
Volume	200+ TB
Growth Rate	~50% annually (estimated)
Data Types	LiDAR, Camera, Radar, CAN, GPS, IMU
Use Cases	Training, Validation, Simulation, Analytics
Infrastructure	Hybrid cloud + On-premise
Pipeline	Real-time + Batch processing

15+ ML Models in Production:

Model Category	Application
Perception	Object detection, Classification
Prediction	Trajectory forecasting
Planning	Path planning, Decision making
Control	Steering, Acceleration, Braking
Safety	Monitor, Fallback systems

SAFe 6 Agilist Certification (2024):

Attribute	Details
Certification	Certified SAFe 6 Agilist
Organization	Scaled Agile, Inc.

Certificate ID	25148732-7562
Validity	October 2024 – October 2025
Application	Program management at scale

Team and Organizational Structure:

Aspect	Scale
Direct Reports	Estimated 5–10 program managers
Cross-functional Teams	50–100+ engineers
Budget Authority	€100M+ (component of €1.9B)
Stakeholder Management	C-level, Board, Partners

Key Challenges & Solutions:

Challenge	Solution Approach
Scale Complexity	SAFe framework implementation
Safety Requirements	ISO 26262, SOTIF compliance
Data Volume	Scalable cloud architecture
Regulatory Uncertainty	Proactive authority engagement
Partnership Coordination	Structured governance models

6.3 Volkswagen Group - Data Product Manager (2018–2020)

Role Transition:

This period marks Fedjaev's transition from academic research to industrial product management—applying data science expertise to automotive applications.

Focus Areas:

Domain	Applications
Big Data Platforms	Connected car data infrastructure
Analytics	Vehicle usage patterns, Customer insights
Data Products	API products, Data services
Cross-brand	VW, Audi, Porsche data integration

Skills Developed:

- **Product Management:** Roadmapping, Prioritization, Stakeholder management
- **Automotive Domain:** Vehicle architectures, ECU communication, OBD
- **Scale Operations:** Enterprise data governance, Security, Compliance
- **Business Acumen:** P&L responsibility, Market analysis

6.4 Current Role - Data Product Lead (2024–Present)

Evolution:

The transition from "Head of Program Management" to "Data Product Lead" reflects:

Strategic Focus: Product strategy over program execution

AI/ML Emphasis: Deepening technical leadership in AI

Broader Scope: Group-level impact beyond autonomous mobility

Current Responsibilities:

Area	Focus
AI/ML Leadership	Data product strategy for autonomous systems
Strategic Alignment	Cross-divisional data product coordination
Innovation	Emerging technology evaluation
Team Development	Product management capability building

7. Consulting & Talent Programs

7.1 Program Portfolio Overview

Fedjaev's participation in multiple top-tier consulting talent programs demonstrates exceptional professional recognition and strategic network development.

Year	Organization	Program	Type	Status
2017	PwC	Next Digital Leader	Talent Development	Alumni
2018	BCG	Emeralds Program	Talent Development	Alumni
2018	McKinsey	#Explore Program	Talent Development	Alumni

2022	McKinsey	Leadership Development	Executive Education	Alumni
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7.2 McKinsey & Company Programs

#Explore Program (2018):

Attribute	Details
Full Name	#Explore - McKinsey's European Operations Practice Program
Year	2018
Practice	European Operations
Focus	Operations, Supply Chain, Manufacturing
Selection	Competitive (estimated <20% admission)
Format	Workshops, Case studies, Networking
Outcome	Alumni network access

Program Value:

- **Methodology:** McKinsey problem-solving approach
- **Network:** Peer cohort of high-potential professionals
- **Exposure:** Client project insights
- **Career:** Pathway to potential McKinsey role

Leadership Development (2022):

Attribute	Details
Program Name	"Inspiration for Tomorrow"
Year	2022
Focus	Executive leadership, Future-oriented thinking
Format	Intensive workshop, Peer learning
Status	Completed

Strategic Significance:

Participation in two McKinsey programs across four years indicates:

Sustained Relationship: Ongoing McKinsey engagement

Leadership Trajectory: Progression from talent to executive development

Network Value: Access to McKinsey ecosystem

7.3 Boston Consulting Group - Emeralds Program

Program Profile:

Attribute	Details
Full Name	BCG Emeralds Program
Year	2018
Type	Talent identification and development
Selection	Highly competitive
Focus	Strategy, Digital transformation

Program Structure (Typical):

- **Assessment:** Case interviews, Problem-solving tests
- **Workshops:** BCG methodology, Strategy frameworks
- **Projects:** Mini-consulting engagements
- **Networking:** Alumni community access

Differentiation:

BCG Emeralds is distinct from general BCG recruitment—it's an exclusive program for pre-identified high-potential talent, often while still in university or early career.

7.4 PwC - Next Digital Leader (2017)

Program Profile:

Attribute	Details
Organization	PricewaterhouseCoopers
Year	2017
Program	Next Digital Leader
Focus	Digital transformation, Technology consulting
Timing	Early career (pre-McKinsey/BCG)

Significance:

This was Fedjaev's first major talent program, likely providing:

- **Consulting Exposure:** First exposure to professional services
- **Digital Focus:** Technology transformation perspective

- **Foundation:** Preparation for subsequent McKinsey/BCG programs

7.5 Program Comparison

Dimension	McKinsey	BCG	PwC
Selectivity	Very High	Very High	High
Focus	Operations/General	Strategy	Digital
Network Quality	Global elite	Global elite	Strong
Career Impact	Significant	Significant	Moderate
Fedjaev Engagement	2 programs	1 program	1 program

7.6 Network Value

Alumni Networks:

Program	Network Size	Value to Fedjaev
McKinsey #Explore	1000s global	Executive network
McKinsey Leadership	100s senior	C-level connections
BCG Emeralds	100s cohort	Strategy network
PwC Next Digital	1000s regional	Technology network

Strategic Applications:

The consulting network provides access to:

- **Best Practices:** Cross-industry methodology exposure
- **Talent Pipeline:** Recruitment and team building
- **Advisory:** Informal strategic advice
- **Partnerships:** Corporate development opportunities

8. Certifications & Education

8.1 SAFe 6 Agilist - Complete Details

Certification: Certified SAFe 6 Agilist

Certificate ID: 25148732-7562

Issuing Organization: Scaled Agile, Inc.

Validity Period: October 2024 – October 2025

Status: Active

SAFe Framework Overview:

The Scaled Agile Framework (SAFe) is the world's most widely adopted enterprise agility framework, designed to address the challenges of scaling agile practices across large organizations.

Aspect	Details
Current Version	SAFe 6.0
Framework Elements	10+ configurations (Essential, Portfolio, Full)
Core Values	Alignment, Built-in quality, Transparency, Program execution
Certification Level	Agilist (Foundational)
Target Roles	Program managers, Change agents, Executives

Curriculum Coverage:

Module	Content
SAFe Principles	10 lean-agile principles
Lean-Agile Mindset	Value stream thinking, Flow optimization
Program Execution	Agile Release Trains (ARTs), PI Planning
Lean Portfolio	Strategy alignment, Investment funding
Leadership	Lean-agile leadership behaviors
DevOps	Continuous delivery pipeline

Career Application:

At VW Autonomous Mobility, SAFe certification enabled:

- **Program Scale:** Managing €1.9B portfolio with agile governance
- **Cross-functional Coordination:** 100+ person ARTs
- **Regulatory Alignment:** Structured documentation for automotive compliance
- **Stakeholder Communication:** Common framework with leadership

8.2 TOEIC Certification

Score: 960/990

Date: January 2015

Level: Professional Proficiency

Context: CDI MBA program requirement

TOEIC Score Interpretation:

Score Range	Level	Description
905–990	Professional	Can communicate effectively in demanding business situations
785–900	Working	Can satisfy most work requirements with language
605–780	Limited	Can satisfy limited work requirements

Fedjaev's Position: 960/990 places him in the top 10% of test-takers globally, demonstrating:

- **Business English:** Presentation, negotiation, documentation
- **Academic English:** Research writing, conference participation
- **Technical English:** Specifications, Architecture documents

8.3 Other Certifications

Based on career trajectory, likely certifications include:

Certification	Probability	Evidence
Product Management	High	Role requirements
Cloud (AWS/Azure)	Medium	Data platform work
Agile/Scrum	High	SAFe prerequisite
Data Engineering	Medium	Platform development

9. Open Source Contributions

9.1 GitHub Portfolio Overview

Fedjaev maintains an active GitHub presence with 23 public repositories spanning research tools, personal projects, and learning materials.

Profile Statistics:

Metric	Value
Member Since	2013 (12+ years)
Public Repositories	23
Followers	15
Following	23
Stars Received	59
Bio	"Data Product Guy at Automotive OEM"
Location	Germany
Website	jfedjaev.de

9.2 Repository Catalog (All 23)

gumpy Organization (gumpy-bci)

#	Repository	Language	Stars	Description
1	gumpy	Python	68	Main BCI toolbox
2	gumpy-deeplearning	Python	32	Deep learning models
3	gumpy-realtime	Python	8	Real-time BCI modules
4	gumpy-online-hybrid-bci	Python	5	Hybrid BCI implementations
5	gumpy-paradigms	Python	3	BCI paradigms collection

Personal Repositories (jfedjaev)

#	Repository	Language	Stars	Description	Year
6	NSTresearchPractice	MATLAB	5	Neuroscience research practice	2017
7	vuejs-project	Vue.js	-	Vue.js learning project	-
8	jfedjaev-portfolio	TypeScript	-	Personal portfolio website	2026

9	namely	Swift	-	iOS name finder app	2025
10	matlab_eeg	MATLAB	-	EEG signal processing	2017
11	P300	MATLAB	-	P300 ERP paradigm	2012
12	python_emotiv	Python	-	Emotiv EPOC driver	2017
13	awesomesauce-rfduino	Java	-	RFduino Android library	2013
14	CppUnrealCourse	C++	-	Unreal Engine course	2017
15–23	Additional repos	Various	-	Various projects	-

9.3 Detailed Repository Analysis

gumpy (68 stars)

README Summary:

gumpy is a Python 3 toolbox for Brain-Computer Interface development. It provides:

- EEG and EMG signal processing
- Feature extraction (temporal, spectral, spatial)
- Classification algorithms
- Visualization tools
- Hybrid BCI support

Dependencies:

```
numpy>=1.14.0
scipy>=1.0.0
scikit-learn>=0.19.0
matplotlib>=2.1.0
```

Installation:

```
pip install gumpy
# or
conda install -c gumpy-bci gumpy
```

Example Usage:

```
import gumpy

# Load EEG data
data = gumpy.data.load('motor_imagery.edf')
```

```
# Preprocess
filtered = gumpy.signal.filter(data, lowcut=8, highcut=30)
features = gumpy.features.csp(filtered)

# Classify
accuracy = gumpy.classification.lda(features, labels)
```

gumpy-deeplearning (32 stars)

Focus: Deep learning models for BCI

Models Included:

- CNN (Convolutional Neural Networks)
- LSTM/GRU (Recurrent architectures)
- CNN-LSTM hybrid
- Attention-based models

Framework: TensorFlow/Keras

Use Case: Research validation (2019 Sensors paper)

NSTresearchPractice (5 stars)

Purpose: Neuroscience research training materials

Content:

- MATLAB tutorials for EEG analysis
- Signal processing exercises
- Visualization examples

Audience: Students, BCI beginners

9.4 Contribution Patterns

Commit Activity (Estimated):

Period	Activity Level	Focus
2012–2016	High	Academic projects, Research
2016–2018	Very High	gumpy development
2018–2020	Moderate	Transition to industry
2020–2024	Low	Corporate role

2024–2026	Moderate	Personal projects (namely, portfolio)
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Code Quality Indicators:

Aspect	Evidence
Documentation	Comprehensive READMEs, gumpy.org website
Testing	Test suites in gumpy
CI/CD	GitHub Actions
Licensing	MIT License (permissive)
Style	PEP 8 compliance

9.5 External Contributions

gumpy Ecosystem:

Fedjaev's contributions extend beyond his own repositories through:

- **Issue Responses:** Community support on GitHub
- **Documentation:** gumpy.org website content
- **Examples:** Tutorial notebooks
- **Citations:** Academic papers using gumpy

10. Awards & Recognition

10.1 Award Portfolio

Year	Award	Organization	Details	Rarity
2014	#MOC2014 Winner	INRIA	"Best Connected Object"	Top 3 projects
2016	Best Paper Award	IEEE PEMC	Battery optimization	~5% of papers
2017	Next Digital Leader	PwC	Talent program	<20% selection
2018	Emeralds Program	BCG	Talent program	<15% selection
2018	#Explore Program	McKinsey	Operations practice	<20% selection

2022	Leadership Development	McKinsey	Executive program	Invitation only
2024	SAFe 6 Agilist	Scaled Agile	Certification	Standard exam

10.2 #MOC2014 Detailed Analysis

Competition: Maison Connectée 2014

Organizer: INRIA (French National Research Institute)

Award: "Best Connected Object"

Project: Tact'Icons 3

INRIA Significance:

INRIA is France's premier computer science research organization, equivalent to:

- Germany: Max Planck Institutes
- US: MIT CSAIL, Stanford HAI
- UK: Alan Turing Institute

Competition Context:

Attribute	Details
Year	2014
Domain	Internet of Things, Connected Home
Participants	Engineering students, Researchers, Startups
Evaluation	Innovation, Technical execution, Viability
Fedjaev Status	Student at Centrale Lille

Impact:

- First major external validation of technical capabilities
- Foundation for IoT career trajectory
- Network entry into INRIA ecosystem

10.3 IEEE Best Paper Award (2016)

Conference: IEEE PEMC (Power Electronics and Motion Control)

Location: Varna, Bulgaria

Paper: Battery optimization for industrial microgrid

Award: Best Paper Award

IEEE PEMC Context:

Attribute	Details
Frequency	Biennial
Scope	Power electronics, Motor control, Energy
IEEE Status	Flagship conference in field
Competition	Hundreds of submissions

Best Paper Selection Criteria:

Technical Quality: Novel contribution, Rigorous methodology

Presentation: Clear communication, Effective slides

Impact: Practical relevance, Field advancement

Q&A Handling: Expert discussion responses

Significance:

Receiving Best Paper at IEEE PEMC as a student indicates:

- Research quality competitive with established academics
- Presentation skills for expert audience
- Cross-domain capability (BCI + Energy)

10.4 Talent Program Prestige**McKinsey #Explore:**

Aspect	Value
Global Reach	European Operations Practice
Selection	Multi-stage interview process
Peers	High-potential professionals across Europe
Outcome	Alumni network, Potential career path

BCG Emeralds:

Aspect	Value
Exclusivity	Invitation-only program
Timing	Typically pre-graduation

Focus	Strategy, Digital transformation
Network	BCG consultant access

Comparison:

Both McKinsey and BCG programs represent top-tier consulting talent identification. Participation in both (plus PwC) creates a "triple crown" of consulting exposure.

11. Alumni Networks & Community

11.1 Network Membership

Organization	Status	Since	Network Size	Benefits
TUM Alumni	Member	2018	70,000+	Career services, Events
CDI Alumni Deutschland	Member	2018	2,000+	Networking, Mentoring
Centrale Lille Alumni	Member	2013	15,000+	Global chapters
TIME Alumni	Member	2013	10,000+	Cross-institutional

11.2 TU Munich Alumni

Institution: Technische Universität München

Joined: 2018 (post-graduation)

Network Scale: 70,000+ alumni globally

Notable Alumni:

- 17 Nobel laureates
- CEOs of BMW, Siemens, Allianz
- Leaders in technology and engineering

Fedjaev Benefits:

- **Career Services:** Job postings, Career counseling
- **Events:** Technical lectures, Networking evenings
- **Mentoring:** Alumni mentorship programs

- **Continuous Learning:** Lecture access, Online courses

11.3 CDI Alumni Deutschland

Institution: Collège des Ingénieurs Germany Chapter

Joined: 2018

Focus: Business-engineering dual qualification network

Network Characteristics:

- **Profile:** Technical executives, Product leaders
- **Industry:** Automotive, Technology, Consulting
- **Activities:** C-level events, Strategy workshops

Fedjaev Fit:

CDI alumni profile aligns closely with Fedjaev's career—technical background with business leadership.

11.4 Centrale Lille Alumni

Institution: Ecole Centrale de Lille

Joined: 2013 (during studies)

Network Scale: 15,000+ alumni

International Presence:

- Paris, London, New York chapters
- Strong in energy, consulting, technology

Fedjaev Connection:

Maintained despite primarily German career—demonstrates commitment to Franco-German professional relationships.

11.5 TIME Alumni

Network: Top Industrial Managers for Europe

Joined: 2013

Member Institutions: 50+ European engineering schools

Unique Value:

- **Cross-Institutional:** Connects alumni from all member schools
- **European Focus:** EU-wide professional network

- **Mobility:** Supports international career moves

12. Speaking & Presentations

12.1 Conference Presentations

IEEE PEMC 2016:

Aspect	Details
Conference	IEEE Power Electronics and Motion Control
Location	Varna, Bulgaria
Presentation	Battery optimization paper
Outcome	Best Paper Award
Audience	Academic and industry researchers

Academic Context:

Conference presentations at IEEE events require:

- **Paper Acceptance:** Peer review (~30% acceptance rate)
- **Presentation Skills:** 15-20 minute technical talk
- **Q&A Competency:** Expert audience questions

12.2 BCI Community Engagement

gumpy Promotion:

Activity	Details
Documentation	gumpy.org website
Tutorials	GitHub examples, YouTube channel
Support	GitHub issue responses
Conference	Workshop presentations (inferred)

YouTube Channel:

- URL: youtube.com/channel/UCdarvfot4Ustk2UCmCp62sw
- Content: gumpy tutorials, BCI demonstrations
- Specific Video: youtu.be/M68GeL8PafE

12.3 Internal Presentations (Inferred)

Volkswagen Autonomous Mobility:

Audience	Topic
Board/Executive	€1.9B program status
Engineering Teams	SAFe implementation, Technical roadmap
Partners	Collaboration frameworks
Regulators	Safety methodology, Compliance

Estimated Presentation Volume:

- **Internal:** 50+ presentations annually (estimated)
- **External:** 5–10 industry events annually (estimated)

13. Media Coverage & Public Presence

13.1 Online Presence Portfolio

Platform	URL	Status	Activity Level
LinkedIn	linkedin.com/in/jfedjaev	Active	High
GitHub	github.com/jfedjaev	Active	Moderate
Website	jfedjaev.de	Active	Low
X/Twitter	x.com/jfedjaev	Linked	Low
Instagram	instagram.com/jfedjaev	Linked	Personal

13.2 LinkedIn Profile Analysis

Current Position: Data Product Lead at Volkswagen Group

Previous: Head of Program Management at VW Autonomous Mobility

Location: Frankfurt, Germany

Network: 500+ connections (estimated)

Profile Strength Indicators:

- Complete work history
- Education verification

- Skill endorsements
- Recommendations

13.3 Personal Website (jfedjaev.de)

Current State:

- Minimalist design
- Links to LinkedIn, X/Twitter, Instagram
- "more to come..." placeholder

Technical Stack:

- Likely static site (Next.js based on portfolio repo)
- Minimal content (intentional or work-in-progress)

Evolution:

The website's minimalism may reflect:

- **Transition Period:** Career pivot to new role
- **Intentional Restraint:** Focus on LinkedIn
- **Development Pending:** New version in progress (portfolio repo suggests update)

13.4 GitHub Presence

Profile: github.com/jfedjaev

Activity Heatmap Pattern:

- **2013–2018:** Consistent activity (academic/research phase)
- **2018–2024:** Reduced activity (corporate role)
- **2024–2026:** Renewed activity (personal projects)

Repository Distribution:

- **Research:** 40% (BCI, EEG)
- **Web/Mobile:** 30% (Portfolio, apps)
- **Learning:** 20% (Courses, tutorials)
- **IoT:** 10% (Tact'Icons)

13.5 Media Coverage Search

Academic Citations:

- 8+ citations of gumpy paper
- 5+ citations of Sensors paper
- IEEE Xplore indexing

Press Coverage:

- No major press articles identified
- Focus on technical/academic channels

Podcast/Interview:

- No public appearances identified
- Potential internal VW communications

14. Tools & Technology Stack

14.1 Development Environment

Primary IDE:

- **Python:** VS Code, PyCharm (inferred)
- **MATLAB:** MATLAB IDE
- **iOS:** Xcode (namely project)
- **Web:** VS Code, Terminal

Version Control:

- **Git:** Primary VCS
- **GitHub:** Primary host (23 repos)
- **Workflow:** Feature branch, Pull requests

Documentation:

- **Python:** Sphinx, Google docstrings
- **General:** Markdown, LaTeX

14.2 Data Science Stack

Core Libraries:

```
# Numerical & Scientific
numpy>=1.14.0 # Array operations
scipy>=1.0.0 # Signal processing
pandas>=0.22.0 # Data manipulation

# Machine Learning
scikit-learn>=0.19.0 # Traditional ML
tensorflow>=2.0.0 # Deep learning
keras>=2.0.0 # Neural network API

# Visualization
matplotlib>=2.1.0 # Publication plots
seaborn>=0.9.0 # Statistical viz
plotly>=4.0.0 # Interactive plots
```

```
# Development
jupyter>=1.0.0 # Notebooks
pytest>=5.0.0 # Testing
black>=20.0.0 # Formatting
```

BCI Specific:

```
# EEG Processing
mne>=0.20.0 # MNE-Python (familiar)
gumpy>=1.0.0 # Own toolbox
pyedflib>=0.1.0 # EDF file format

# Signal Processing
pywavelets>=1.0.0 # Wavelet transforms
```

14.3 Web Development Stack

Frontend:

```
// Frameworks
vue@3.x // Progressive framework
react@18.x // UI library
next@14.x // React framework

// Styling
tailwindcss@3.x // Utility-first CSS

// Build
vite@5.x // Fast dev server
webpack@5.x // Module bundler
```

Backend (Inferred):

```
node@18.x // Runtime
express@4.x // Web framework
```

14.4 Mobile Development

iOS:

```
swift@5.x // Language
xcode@15.x // IDE
swiftui@5.x // UI framework (inferred)
```

Android (Historical):

```
java@8.x // Language
android-sdk@28.x // Platform
```

14.5 Infrastructure & DevOps

Cloud Platforms:

- AWS (inferred from VW context)
- Azure (inferred)
- Google Cloud (inferred)

Containerization:

- Docker
- Kubernetes (inferred)

CI/CD:

- GitHub Actions
- Jenkins (inferred)

14.6 Favorite Tools (Inferred)

Category	Tool	Rationale
Note-taking	Notion/Obsidian	Knowledge management
Diagramming	draw.io/Excalidraw	Architecture diagrams
Communication	Slack/Teams	Corporate standard
Project Management	Jira/Linear	SAFe implementation
Data Exploration	Jupyter Lab	Interactive analysis
Version Control	GitHub	Long-term usage

15. Future Vision & Goals

15.1 Career Trajectory Analysis

Historical Pattern:

Phase	Focus	Outcome
2010–2016	Academic Foundation	TUM Master, Publications
2016–2018	Research Excellence	gumpy, IEEE papers
2018–2020	Industry Transition	VW Data PM
2020–2024	Program Leadership	€1.9B programs, SAFe
2024+	Strategic Leadership	Data Product Lead

Trajectory Indicators:

- Increasing scope: Research → Product → Program → Strategy
- Growing scale: Papers → Tools → Platforms → Portfolios
- Expanding influence: Academic → Team → Division → Group

15.2 Likely Career Directions

Scenario 1: Automotive Leadership

- **Path:** VP Product → Chief Product Officer (OEM)
- **Focus:** Autonomous driving product strategy
- **Timeline:** 5–10 years
- **Probability:** High

Scenario 2: Mobility Entrepreneurship

- **Path:** Startup founder (AD tech, BCI applications)
- **Focus:** Next-generation mobility interfaces
- **Timeline:** 2–5 years
- **Probability:** Medium

Scenario 3: Investment/Advisory

- **Path:** VC partner, Mobility-focused fund
- **Focus:** AD tech investing
- **Timeline:** 5+ years
- **Probability:** Medium

Scenario 4: BCI Industry Application

- **Path:** Neurotechnology industry leadership
- **Focus:** Automotive-Human Interface
- **Timeline:** Variable
- **Probability:** Low-Medium

15.3 Strategic Interests

Emerging Technology Focus:

Domain	Interest Level	Application
Generative AI	High	AD simulation, Synthetic data
Neurotechnology	Maintained	Driver monitoring, HMI
Quantum Computing	Curious	Optimization problems
Sustainable Mobility	High	EV strategy, Carbon reduction
Urban Air Mobility	Interested	Regulatory, Safety

Current Project Pipeline:

Project	Status	Description
jfedjaev-portfolio	Active	Personal website update

namely	Active	iOS app development
BCI Research	Maintenance	gumpy community support

15.4 Development Goals

Technical:

- Maintain Python/MATLAB expertise
- Expand Swift/mobile capabilities
- Stay current with GenAI developments

Leadership:

- Scale team and organizational impact
- Develop next-generation product leaders
- Contribute to industry standards

Network:

- Leverage consulting alumni relationships
- Expand neurotechnology connections
- Build mobility executive network

Appendices

Appendix A: Source Documentation

Primary Sources

#	Source	URL	Date Verified
1	LinkedIn	linkedin.com/in/jfedjaev	2026-01-30
2	GitHub Personal	github.com/jfedjaev	2026-01-30
3	GitHub gumpy	github.com/gumpy-bci/gumpy	2026-01-30
4	Personal Website	jfedjaev.de	2026-01-30
5	X/Twitter	x.com/jfedjaev	2026-01-30
6	Instagram	instagram.com/jfedjaev	2026-01-30

Academic Sources

#	Publication	DOI	Journal
1	gumpy (2018)	10.1088/1741-2552/aae186	Journal of Neural Engineering
2	Deep Neural Networks (2019)	10.3390/s19010210	Sensors (MDPI)
3	Battery Optimization (2016)	10.1109/PEMC.2016.7752032	IEEE PEMC
4	Master Thesis (2017)	mediatum.ub.tum.de/doc/1422453	TU München

Institutional Sources

#	Institution	URL
1	TU München	tum.de
2	Ecole Centrale Lille	centraledille.fr
3	Collège des Ingénieurs	cdi.eu
4	Volkswagen Group	volkswagenag.com
5	McKinsey	mckinsey.com
6	BCG	bcg.com
7	PwC	pwc.com
8	Scaled Agile	scaledagileframework.com

Appendix B: Research Methodology

Data Collection:

- Web scraping (public profiles)
- API queries (GitHub)
- Academic database search (DOI resolution)
- Cross-reference verification

Limitations:

- Private information not accessible
- Some details inferred from context
- Corporate role specifics confidential

Verification:

- Multi-source confirmation where possible
- DOI verification for publications

- Timestamp checking for consistency

Appendix C: Glossary

Term	Definition
AD	Autonomous Driving
ART	Agile Release Train (SAFe)
BCI	Brain-Computer Interface
CDI	Collège des Ingénieurs
CSP	Common Spatial Patterns
ECG	Electrocardiogram
EEG	Electroencephalography
ERP	Event-Related Potential
GRU	Gated Recurrent Unit
IoT	Internet of Things
LSL	Lab Streaming Layer
LSTM	Long Short-Term Memory
MLOps	Machine Learning Operations
OEM	Original Equipment Manufacturer
P300	Positive 300ms ERP component
SAFe	Scaled Agile Framework
TIME	Top Industrial Managers for Europe
TUM	Technische Universität München

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End of Report